

Exam Date & Time: 02-Dec-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov / Dec -2024

Social Networks [NPT-3204]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Using the following Python code snippet, answer the questions below:
- ```
import networkx as nx
G = nx.Graph()
A) G.add_nodes_from(["Alice", "Bob", "Charlie", "David", "Eve"])
G.add_edges_from([("Alice", "Bob"), ("Alice", "Charlie"), ("Bob", "David"), ("Charlie", "Eve")])
degree centrality = nx.degree_centrality(G)
```
1. What does the function `nx.degree_centrality(G)` compute, and how would it be useful in analyzing the social network of this community? (5)
  2. Calculate the degree centrality of each node in the network. Which individual has the highest degree centrality, and what does this imply about their position in the network?
- If the edge ("David", "Eve") is added to the graph, how will the degree centrality of each individual change?
- B) Gephi offers various layout algorithms (e.g., ForceAtlas 2, Fruchterman-Reingold). How do these algorithms help in visualizing network structures, and what are the advantages of using force-directed layouts in social network analysis? (3)
- C) How do you decide which centrality measure (degree, betweenness, closeness, eigenvector) to use for identifying key nodes in a network? Are there cases where a node with low degree centrality could be more important based on another centrality measure? Provide an example to illustrate this. (2)
- 2) (5)
- A)

Consider the following scenario in a small professional network of 6 individuals:

**Network:**

- **Nodes (Individuals):** Alice, Bob, Charlie, David, Eve, Frank
- **Edges (Friendship/Professional connections):**
  - Alice - Bob
  - Alice - Charlie
  - Bob - David
  - Bob - Eve
  - Charlie - Eve
  - David - Frank

**Interactions and Context:**

- **Alice and Bob:** They work closely together in the same department and meet for coffee frequently. They often discuss work-related ideas and personal matters.
- **Alice and Charlie:** They occasionally attend the same work meetings but don't communicate outside of those meetings.
- **Bob and David:** Bob and David collaborate on several large projects but rarely meet in person, mostly communicating through email.
- **Bob and Eve:** Bob and Eve share some interests outside of work and occasionally meet socially, though they rarely discuss work.
- **Charlie and Eve:** They are colleagues in different departments and meet at annual company-wide events but don't interact often.
- **David and Frank:** David and Frank met at a professional conference, and they keep in touch sporadically over LinkedIn, but they don't have an ongoing work relationship.

**Questions:**

1. **Identify Strong Ties:**
  - Which individuals in this network have **strong ties**? Explain the basis for your identification, considering the frequency and nature of their interactions.
2. **Identify Weak Ties:**

Which individuals have **weak ties**? Justify your reasoning with reference to the frequency, depth, and context of their interactions.

- B) Consider a social network where we have a small subset of nodes: **A, B, C, and D**. The relationships (edges) between them are as follows:

- **A is connected to B and C.**
- **B is connected to C and D.**
- **C is connected to D.**

(3)

**Answer below questions**

1. Identify all the triads in this network.
2. Classify each triad as **open** or **closed**

- C) In a network of 1,000 people, if a highly influential individual starts promoting a product (2) and influences 10 people in the first round, and each influenced person subsequently influences 5 new people in the next round, how many people will have been influenced

after two rounds?

- 3) How does spatial segregation in social networks contribute to economic inequality? Describe the feedback loop between segregated communities and unequal access to educational, employment, and social opportunities, providing examples from real-world scenarios. (5)
- A)
- B) In a social network of **6 people** (P, Q, R, S, T, U), there are **10 positive relationships** and **5 negative relationships**. Each person has relationships with at least 3 others.  
**a)** If you were to convert one negative relationship into a positive one to increase overall network harmony, how many possible combinations would there be for choosing one negative relationship to change? (3)  
**b)** After changing one negative relationship, calculate the new ratio of positive to negative relationships.
- C) How might increasing positive relationships around a highly central (influential) node help stabilize the entire network? (2)
- 4) While power-law distributions are often used to describe large-scale social networks, there are criticisms and challenges associated with their application. Discuss the limitations of using power-law models to describe real-world networks. What are the key criticisms related to the fit of the power-law model to network data, and how might researchers address these concerns when analyzing real-world social networks? (5)
- A)
- B) Examine how does the "richer gets richer" effect (preferential attachment) influence the degree distribution in a social network, and what does it mean for the emergence of highly influential nodes? (3)
- C) Identify the key characteristics of an Erdos-Renyi(ER) network, particularly in terms of degree distribution and clustering. How does the ER model differ from real-world social networks in these aspects? (2)
- 5) Discuss the mathematical foundations behind the small world model, and explore its real-world applications, such as the "six degrees of separation" theory, viral marketing, and information diffusion. (5)
- A)
- B) In the context of **information diffusion**, how does the **cascading model** explain the process through which information spreads from one user to others in a network? (3)
- C) Mention the significance of **short path lengths** in a social network for the rapid spread of information or viral content? (2)

-----End-----